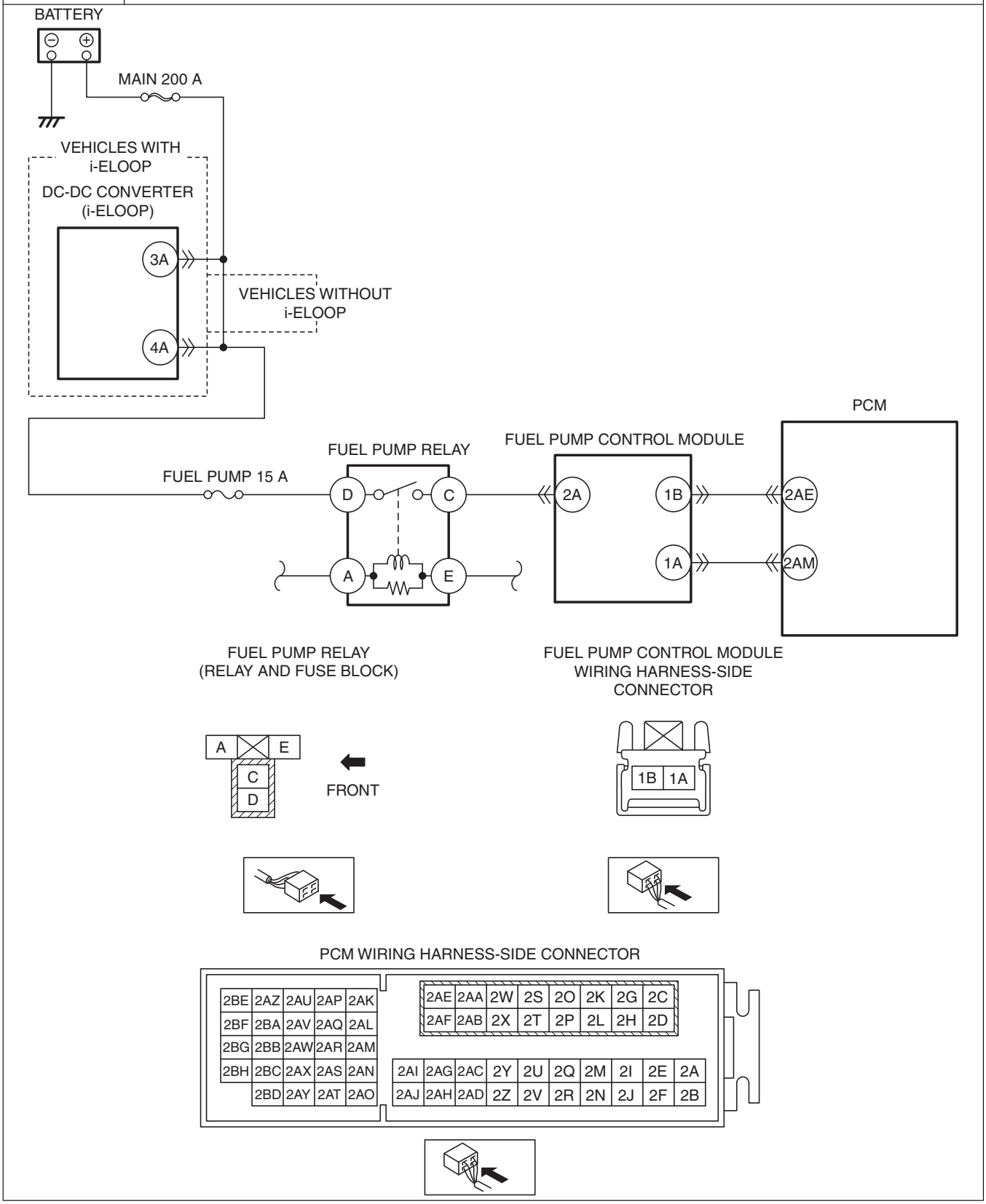
DTC P025D:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

id0102q1008500

DTC P025D:00	Fuel pump control module result of on-board diagnostic test high input
DETECTION CONDITION	• Result of on-board test cannot be received from fuel pump control module (voltage is too high). **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light does not illuminate. • FREEZE FRAME DATA is not available. • Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Fuel pump control module connector or terminals malfunction • PCM connector or terminals malfunction • Short to power supply in wiring harness between fuel pump control module terminal 1A and PCM terminal 2AM • Open circuit in wiring harness between fuel pump control module terminal 1A and PCM terminal 2AM • Fuel pump control module malfunction • Short to ground in wiring harness between fuel pump relay terminal C and fuel pump control module terminal 2A • Short to power supply in wiring harness between fuel pump relay terminal C and fuel pump control module terminal 2A • Open circuit in wiring harness between fuel pump relay terminal C and fuel pump control module terminal 2A • Fuel pump relay malfunction • Short to ground or open circuit in fuel pump relay power supply circuit (without i-ELOOP) — Short to ground in wiring harness between battery positive terminal and fuel pump relay terminal D — MAIN 200 A fuse and/or FUEL PUMP 15 A fuse malfunction — Open circuit in wiring harness between battery positive terminal and fuel pump relay terminal D • Short to ground or open circuit in fuel pump relay power supply circuit (with i-ELOOP) — Short to ground in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and fuel pump relay terminal D — FUEL PUMP 15 A fuse malfunction — Open circuit in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and fuel pump relay terminal D • PCM malfunction

DTC P025D: 00	Fuel pump control module result of on-board diagnostic test high input



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT FUEL PUMP CONTROL MODULE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the fuel pump control module connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 14.
		No	Go to the next step.
4	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 14.
		No	Go to the next step.
5	**INSPECT FUEL PUMP CONTROL MODULE SIGNAL CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the fuel pump control module and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the fuel pump control module terminal 1A (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between fuel pump control module terminal 1A and PCM terminal 2AM. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 14.
6	**INSPECT FUEL PUMP CONTROL MODULE SIGNAL CIRCUIT FOR OPEN CIRCUIT** • Verify that the fuel pump control module and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between fuel pump control module terminal 1A (wiring harness-side) and PCM terminal 2AM (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between fuel pump control module terminal 1A and PCM terminal 2AM. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 14.
7	**INSPECT FUEL PUMP CONTROL MODULE** • Reconnect all disconnected connectors. • Inspect the fuel pump control module. (See FUEL PUMP CONTROL MODULE INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Replace the fuel pump control module, then go to Step 14. (See FUEL PUMP CONTROL MODULE REMOVAL/ INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
8	**INSPECT FUEL PUMP RELAY CONTROL CIRCUIT FOR SHORT TO GROUND** • Remove the fuel pump relay. • Verify that the fuel pump control module connector is disconnected. • Inspect for continuity between fuel pump relay terminal C (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between fuel pump relay terminal C and fuel pump control module terminal 2A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 14.
		No	Go to the next step.
9	**INSPECT FUEL PUMP RELAY CONTROL CIRCUIT FOR SHORT TO POWER SUPPLY** • Fuel pump relay is removed. • Verify that the fuel pump control module connector is disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the fuel pump relay terminal C (wiring harness-side). • Is the voltage 0 V?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between fuel pump relay terminal C and fuel pump control module terminal 2A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 14.
10	**INSPECT FUEL PUMP RELAY CONTROL CIRCUIT FOR OPEN CIRCUIT** • Fuel pump relay is removed. • Verify that the fuel pump control module connector is disconnected. • Switch the ignition off. • Inspect for continuity between fuel pump relay terminal C (wiring harness-side) and Fuel pump control module terminal 2A (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between fuel pump control module terminal 2A and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Inspect for the following: — Open circuit between fuel pump control module and body ground — Loose or lifting ground point • Repair or replace the malfunctioning part. Go to Step 14.
11	**INSPECT FUEL PUMP RELAY** • Switch the ignition off. • Fuel pump relay is removed. (See RELAY LOCATION.) • Inspect the fuel pump relay. (See RELAY INSPECTION.) • Is there any malfunction?	Yes	Replace the fuel pump relay, then go to Step 14.
		No	Vehicles without i-ELOOP: • Go to the next step. Vehicles with i-ELOOP: • Go to Step 13.

STEP	INSPECTION		ACTION
12	**INSPECT FUEL PUMP RELAY POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Fuel pump relay is removed. • Verify that the fuel pump control module connector is disconnected. • Measure the voltage at the fuel pump relay terminal D (wiring harness-side). • Is the voltage B+?	Yes	Go to Step 14.
		No	Inspect the MAIN 200 A fuse and FUEL PUMP 15 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and fuel pump relay terminal D. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the malfunctioning fuse. • If the fuse is damaged: — Replace the malfunctioning fuse. • If all fuses are normal: — Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and fuel pump relay terminal D. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 14.

STEP	INSPECTION		ACTION
13	**INSPECT FUEL PUMP RELAY POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Fuel pump relay is removed. • Verify that the fuel pump control module connector is disconnected. • Measure the voltage at the fuel pump relay terminal D (wiring harness-side). • Is the voltage B+?	Yes	Go to the next step.
		No	Inspect the FUEL PUMP 15 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and fuel pump relay terminal D. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the malfunctioning fuse. • If the fuse is damaged: — Replace the malfunctioning fuse. • If all fuses are normal: — Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and fuel pump relay terminal D. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
14	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Perform the KOEO or KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
15	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.